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ORIGINAL PAPER

Proposing an ensemble learning model based on neural network and fuzzy system for keratoconus diagnosis based on Pentacam measurements

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Abstract
Purpose The present study was done to evaluate efficiency of an ensemble learning structure for automatic keratoconus diagnosis and to categorize eyes into four different groups based on a combination of 19 parameters obtained from Pentacam measurements.
Methods Pentacam data from 450 eyes were enrolled in the study. Eyes were separated into training, validation, and testing sets. An ensemble system was used to analyze corneal measurements and categorize the eyes into four groups. The ensemble system was trained to consider indices from both anterior and posterior corneal surfaces. Efficiency of the ensemble system was evaluated and compared in each group.
Results The best accuracy was achieved by the ensemble system with both multilayer perceptron and neuro-fuzzy system classifiers alongside the Naïve Bayes combination method. The accuracy achieved in KC versus N distinction task was equal to 98.2% with 99.1% of sensitivity and 96.2% of specificity for KC detection. The global accuracy was equal to 98.2% for classification of 4 groups, with an average sensitivity of 98.5% and specificity of 99.4%.
Conclusion In this study, authority of an ensemble learning system to work out intricate problems was presented. Despite using fewer parameters, herein, comparable or, in some cases, better results were obtained than methods reported in the literature. The proposed method demonstrated very good accuracy in discriminating between normal eyes and different stages of keratoconus eyes. In some cases, it was not possible to directly compare our results with the literature, due to differences in definitions of KC group as well as differences in selection of items and parameters.

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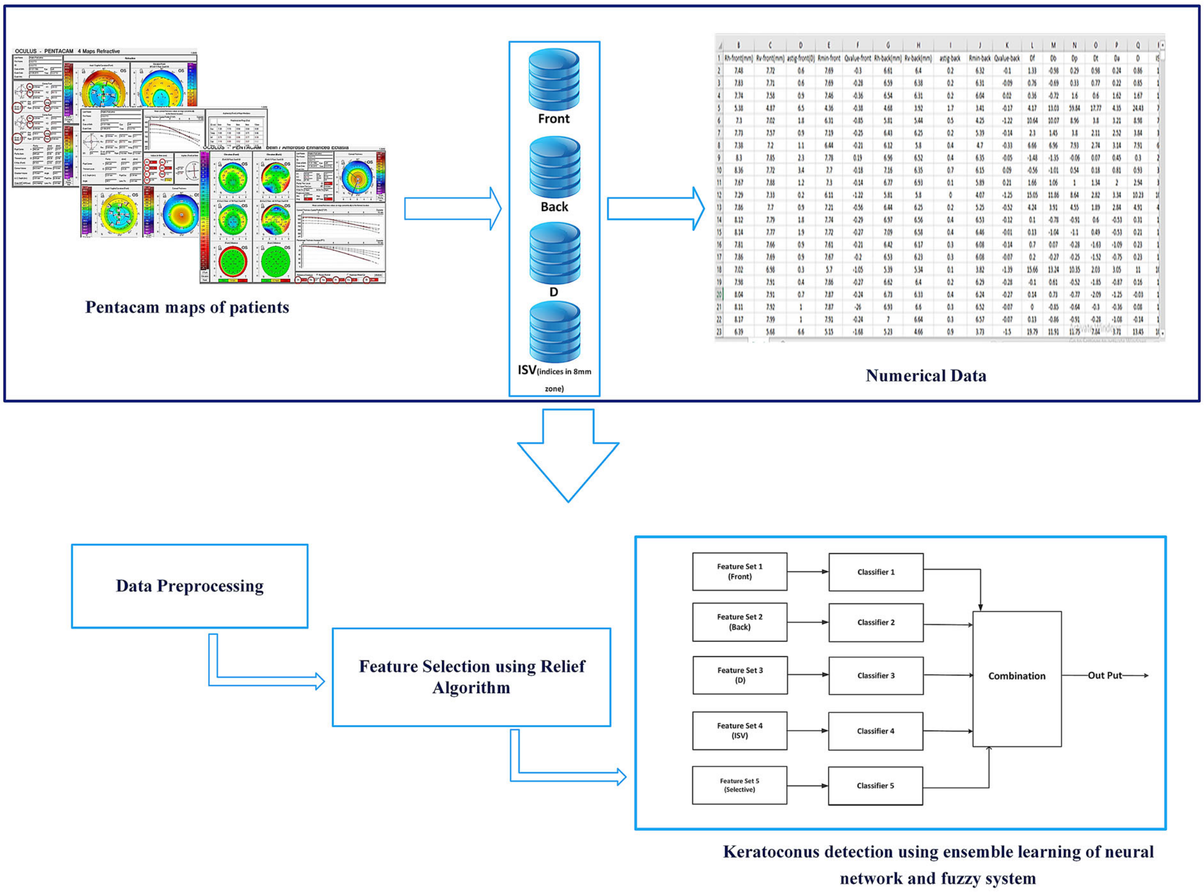
Graphic abstract



Keywords Ensemble learning · Keratoconus · Pentacam · Artificial neural network · Neuro-fuzzy · Machine learning

Introduction

Keratoconus (KC) is a progressive eye disease characterized by a localized conic protrusion with stromal thinning. Misbalance of enzymes in cornea results in this complication, making cornea more sensitive, causing it to become attenuated and protuberated forward [1]. When light enters the eye, this conical shape reverberates light and leads to distorted vision [2]. Increased irregularity in cornea shape causes irregular astigmatism, the advanced nearsightedness, blurred vision, and other problems [2, 3].

KC can occur in one or both eyes and often becomes obvious at early 20 s of pubescence. It can proceed gradually, fixing over almost 10 years, or quickly, and in 10–25% of cases, keratoplasty needs to be performed [4]. KC is categorized into four stages of early (KC1), moderate (KC2), advanced (KC3), and severe (KC4). Each stage of KC requires its different specific treatment steps, ranging from applying implantable ring and contact lenses to partial or full-thickness corneal transplants [5]. Therefore, it is very important to diagnose progression of the disease and know about its stage.

Optometrists usually diagnose the possibility of KC by observing some pathognomonic signs in one's cornea [6]. Glasses or soft contact lenses may assist the patient in the moderate KC. However, when the disease proceeds, glasses and contact lens no longer supply sufficient vision correction. Corneal topography is a valuable diagnostic tool for diagnosing KC and tracking its progression [2, 3].

At present, computer-aided videokeratoscopes, producing colorful maps and topographic indices are the most precise and advanced systems for corroborating diagnosis of KC as well as tracking its progression [1, 7]. Pentacam and Orbscan are methods by which thickness of areas 8–10 mm from center of the cornea can be measured. The images taken by Pentacam are more accurate and clearer than the previous systems. Furthermore, Pentacam is able to measure many parameters of surface and depth of the cornea, which cannot be obtained by ordinary topographies [8, 9]. Application of this technology can help specialists to have a correct judgment. However, despite using this device, there are controversies among specialists in the field of cornea regarding diagnosis of this disease. As such, it is essential to analyze the data obtained from this device [8, 10].

The information obtained by Pentacam arises from different parts of the cornea, which includes a set of numbers. The problem with Pentacam is its inability to compare the relationship and correlation between the information. While, physicians understand condition of the cornea based on this information, as well as their experience and knowledge. Topographic maps of eyes with KC show a different pattern [11] that may be mistaken with other corneal disorders or some cases that do not belong to a certain stage of KC. These cases are suspected of belonging to two adjacent stages. For example, they may belong to the group 3 or 2, or belong to the group 2 or 1, and so on. These samples are classified as suspicious samples. Thus, sometimes, uncertainties persist regarding right diagnosis of KC [4, 12, 13].

In cases of explanation of complicated map, numeric methods have been used to aid the clinician in diagnosis of KC [6, 14]. Considerable attempts and advancements have been made in image processing for cornea and retina through data mining and machine learning techniques to present automatic systems in order to diagnose different disorders. For example, Tetsuro et al. investigated and analyzed videokeratography data to evaluate conditions of cornea and astigmatism [15]; and they assessed irregular astigmatism via harmonic analysis of videokeratography data by Fourier series. In 2014, Martin-Guerre et al. started an investigation on KC. They proposed a new method to predict the increase in patients' sight, who suffered from KC and astigmatism, after ring implantation. Their proposed method evaluated the increase in sight via corneal curvature and astigmatism [16]. In this study, it is attempted to detect various corneal patterns based on Pentacam tomography using a machine learning algorithm. The method, alongside other clinical data, will be able to aid the ophthalmologist in diagnosis of patients with KC.

In this regard, using data mining techniques, the information obtained from Pentacam was compared, and the relationships and correlations between them were investigated. The main purpose of this study is evaluating efficiency of an ensemble learning structure for automatic diagnosis of KC. Ensemble systems are based on combining multiple initial classifiers as experts for primary classification and a combination rule for combining results of classifiers.

Design of this system includes four stages: preprocessing, creating base classifiers, designing combination method, and evaluating system results. The multilayer perceptron (MLP) neural network and an adaptive network-based fuzzy inference system (ANFIS) were used in this study as base classifiers. Naive Bayes and weighted voting combination methods were applied to combine results of these experts.

The rest of this paper is organized as follows: 'Materials and Methods' Section comprises descriptions of basic concepts, such as MLP neural network, ANFIS, and combination methods. A complete description of the proposed model is provided in 'Proposed Method' Section. A summary of data collection, data pre-processing, and evaluation criteria of classification is provided in 'Statistical Population and Dataset' Section. The results obtained from the dataset, as well as discussion about findings and conclusions are expressed in the last sections.

Materials and methods

This section briefly describes some basic concepts, such as ensemble learning, (MLP) neural network, and ANFIS, used as initial classifiers and also, Naive Bayes and weighted voting combination methods used to combine results of classifiers.

Ensemble learning

Classification is a process where an unknown pattern is assigned to one of the unknown classes based on its characteristics [17, 18]. In this study, using MATLAB software 2014b, a method is proposed to diagnose KC via ensemble learning. Ensemble systems employ several learning algorithms to gain better performance than any of classifiers alone [19, 20]. One of the most important conditions in ensemble systems is observing diversity of initial classifiers [21, 22]. Classifiers can be combined in two different ways: selection and fusion [23]. In selection, each part of the problem space is investigated by one of classifiers. The final decision is determined by aggregating decisions of one or some classifiers [20, 22], while in fusion, all classifiers are trained individually on the whole problem space. The decision of all classifiers would determine the final decision [22, 23]. Given type of KC data, herein, selection strategy was used.

Initial classifiers

Multilayer perceptron neural network

Artificial neural networks are designed to identify patterns. These networks consist of several algorithms to help us cluster and classify data based on resemblances between inputs. Moreover, they require a labeled dataset to be trained [24].

In this system, MLP neural networks are used with 3 layers and the backpropagation algorithm [25, 26]. The MLP networks used in this study are composed of one output layer, one hidden layer, and one input layer. Activation function of the hidden layer and that of the output layer was tangent sigmoid (tansig) [26]. Various numbers of neuron in the hidden layer, an adequate number of iterations, and various learning rates were considered to ensure about network results and figure out which network and with what parameters can diagnose KC better. Finally, a MLP with 10 neurons in its hidden layer was selected.

Neuro-fuzzy systems

The ANFIS used in this system involved implementing a Takagi–Sugeno fuzzy system based on a network-like structure [27, 28]. Since this system integrates both neural networks and fuzzy logic principles, it has the potential to capture benefits of both in a single framework. This system has a high level of flexibility and accounting efficiency suggesting complicated and high levels of modeling and estimation [29, 30]. The network had four layers. The first layer was the input layer that included membership functions. In this system, three Gaussian membership functions were used for each input in the neuro-fuzzy network. Rules were formed in the second layer, where a multiplication function was used to create the fuzzy rules. Nodes are known as rule nodes. The third layer was deactivation layer; the final output was obtained from sum of outputs of the fourth layer [31, 32].

Combination methods

Weighted voting

One of the simplest and most common methods for combining different classifiers, which does not treat equally, is the weighted voting combination method. Suppose that N experts are used in the ensemble system. Each of these experts produces a single output for the input sample [33]. Suppose the DA dataset is a five-class dataset, which is applied to assess each classifier. Matrix W (an $N \times M$ matrix) presents efficiency of each classifier Ci (i = 1, 2,…, N) in evaluating the DA dataset.

$$\boldsymbol{W} = \begin{bmatrix} \boldsymbol{W}_{1,1} & \boldsymbol{W}_{1,2} & \ldots & \boldsymbol{W}_{1,N} \\ \boldsymbol{W}_{2,1} & \boldsymbol{W}_{2,2} & \ldots & \boldsymbol{W}_{2,N} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{W}_{N,1} & \boldsymbol{W}_{N,2} & \ldots & \boldsymbol{W}_{N,M} \end{bmatrix} \tag{1}$$

where each element $w_{i,j}$ is defined by

$$2W_{2i,2j} = \frac{22p_{2j}^{(2C_{2i})}}{|\mathbf{DA}_{2j}| + 2p_{2j}^{(2C_{2i})} + 2q_{2j}^{(2C_{2i})'}} \tag{2}$$

Equation (2) evaluates performance of all classifiers. Such that, $W_{i,j}$ is the F1-score of C_i classifier for class j [34]. In this equation, $p_j^{(C_i)}$ and $q_j^{(C_i)}$ are the total number of correct and wrong predictions of Ci classifier, respectively. DA j is the set of samples, belonging to class j. The following equation predicts that each unknown sample of the dataset belongs to which class.

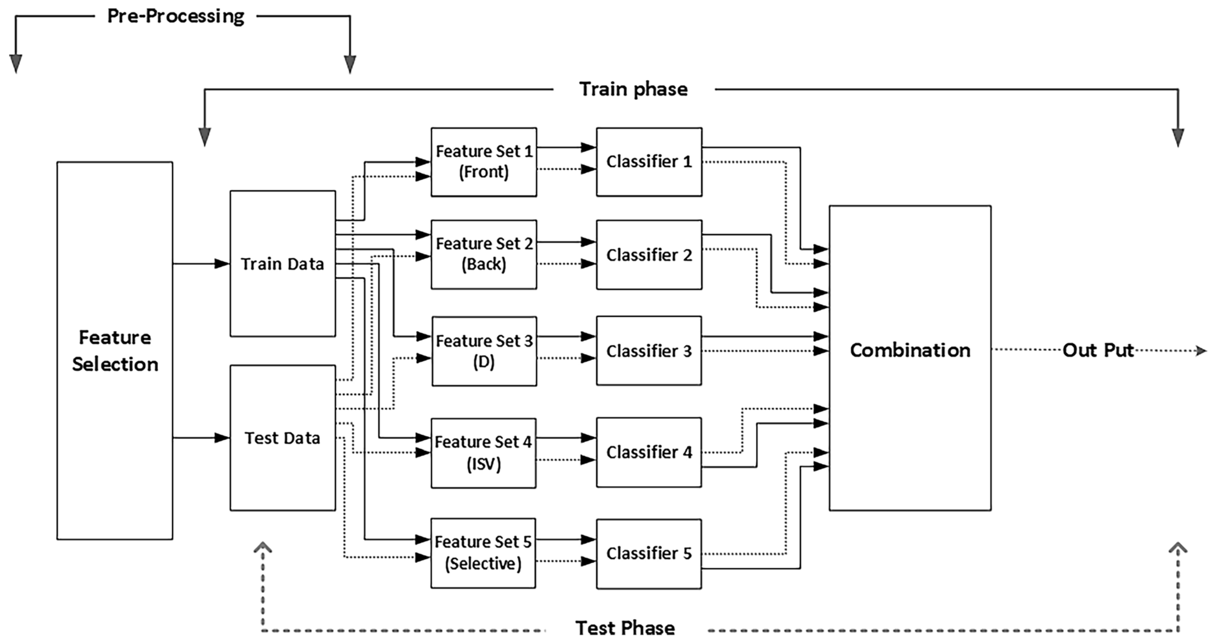


Fig. 1 Structure of our ensemble model. It contains pre-processing phase, five initial classifiers, and a combination method (The figure was created using Microsoft Visio Drawing software)

$$y = \arg\max \sum\nolimits_{i=1}^{N} W_{i,j}\chi_A(C_i(x) = j) \quad (3)$$

In this equation, χ_A is a function, forming a vector based on prediction of $j \in A$ of classifier ci on sample X. In this vector, the j coordinate takes a value of one, and the rest of them take the value of zero. Value of index corresponding to the greatest value of an array is returned by the arg max function [33, 34].

Naive Bayes

Naïve Bayes as a classification method is based on Bayes' Theorem. These classifiers are a family of simple 'probabilistic classifiers' assuming strong (naïve) independence between features. In other words, a Naive Bayes classifier supposes that the presence of a specific feature in a class is unrelated to the presence of any other feature [35].

Suppose the system has N base classifiers as expert, '$D_{j..}$ A dataset 'DA with M classes is applied for evaluation of each classifier. A confusion matrix (cm) is formed for each classifier. Value of cm_j (k, s) presents the number of samples belonging to the class k, but the D_j classifier assigns them to the class s. The parameter cm_j (s) represents the number of samples assigned to the class s. This parameter is sum of the samples of column s of the confusion matrix [17].

$$\boldsymbol{cm^j}(\boldsymbol{s}) = \sum_{\boldsymbol{k=1}}^{\boldsymbol{k}} \boldsymbol{cm^j}(\boldsymbol{k.s}) \quad (4)$$

Using the cm_j (s) and cm_j (k, s), the label matrix (LM) can be computed. Value of Lm_j (k, s) is determined as follows:

$$\boldsymbol{Lm^j}(\boldsymbol{k},\boldsymbol{s}) = \boldsymbol{p}\left(\boldsymbol{w_k}|\boldsymbol{D_j}(\boldsymbol{x}) = \boldsymbol{w_s}\right) = \frac{\boldsymbol{cm^j}(\boldsymbol{k.s})}{\boldsymbol{cm^j}(\boldsymbol{s})} \quad (5)$$

If S_1, S_2, …, S_L are the labels assigned to sample x by classifiers D_1 *to* D_L, the probability that actual label of sample x is w_i is calculated as follows:

$$\mu_D^i(\boldsymbol{x}) = \prod_{j=1}^{1} \boldsymbol{p}\left(\boldsymbol{w_i}|\boldsymbol{D_j}(\boldsymbol{x}) = \boldsymbol{S_j}\right) = \prod_{j=1}^{1} \boldsymbol{Lm^j_{i,s}}\ \boldsymbol{i} = \boldsymbol{1},\ldots,\boldsymbol{k} \quad (6)$$

Proposed method

The proposed ensemble system is introduced in this section. Figure 1 displays the final design of the proposed model. As observed in Fig. 1, design of this system involves three phases: pre-processing phase, training phase, and testing phase. Data normalization and feature selection are done in pre-processing phase. The training phase contains two steps: construction of initial classifiers (MLP or ANFIS) and design of combining method. Initial datasets are in four separate groups of different indices from both anterior and posterior corneal surfaces. Furthermore, one group with the most valuable indices of each group, obtained from feature selection step, was formed as the fifth group. Then, the classifiers were trained over training samples for the initial classification. The results of classifications were combined using one of combining methods. The system results could approach actual values by repeating this process and improving base classifiers at each of iterations. Finally, evaluation of the system was done based on results of the testing data.

Statistical population and dataset

This study is a cross-sectional and retrospective observational analysis. The data used in this study were collected and analyzed from medical records of patients who had visited the Noor Medical Eye Clinic and had been examined there over a period of 8 months. This section provides a brief description of data collection, data pre-processing, and evaluation criteria of classification.

Data collection

For simulation of process of KC diagnosis, a dataset of ‘450 eyes (KC dataset) was used, provided by the research group of this study in Noor Medical Eye Clinic. The patients had keratoconus in both eyes, but one of the eyes was randomly included in the study. The dataset consisted of information from both anterior and posterior corneal surfaces. Since highly irregular shape of the cornea would be notable in topographic maps, severe stage of KC (KC4) is not usually a diagnostic challenge. Thus, in this study, it is tried to investigate early, moderate, and advanced KC (KC types 1–3), which are greater diagnostic challenges. Accordingly, the eyes, studied in this research, were divided into four classes based on the topography results of cornea center.

A skilled operator carried out Scheimpflug tomography for all patients using the Pentacam *HR* (*Oculus, Wetzlar, Germany*). Although the Pentacam software can provide over 1000 parameters, after consulting with several experts, key and important parameters in diagnosis of KC and only those that are available in the most basic version of the Pentacam software were used to facilitate independent replication of the study. Since, this set still comprises of irrelevant or highly correlated variables, adding redundant information, it is essential to lessen its size further to simplify both computation and interpretation of the classifiers. For this purpose, the Relief-based feature selection method was performed [36]. This procedure led to generation of a dataset of 19 variables (RH,[1] RV,[2] Astig,[3] *Q*-value[4] and *R*-Min[5] indices for both anterior and posterior corneal surfaces; Belin / Ambrosio indices (Df,[6] Dp,[7] Dt,[8] Da[9]), and refractive indices in 8 mm zone (ISV,[10] IVA,[11] KI,[12] CKI,[13] IHD[14])). These indices were categorized in four different groups (Front, Back, D and ISV). The chosen indices included aspects of

[1] Horizontal Radius: Horizontal curvature radius of 3 mm cornea in millimeters.

[2] Vertical Radius: Vertical curvature radius of 3 mm of the cornea in millimeters.

[3] Astigmatism: Determines the amount of corneal anterior astigmatism.

[4] Corneal aspheric index is usually reported for a 6 mm central ring.

[5] Minimum Radius: The minimum radius of the anterior surface curvature in millimeters is determined.

[6] Front elevation: The standard deviation from the mean change in the anterior elevation.

[7] Pachymetric Progression: The standard deviation of the mean progression (sequence) of measuring the thickness of the cornea.

[8] Thinnest Point: Standard deviation from the average thickness of the thinnest point.

[9] Thinnest Point Displacement: The standard deviation from the mean vertical displacement of the thinnest point relative to the vertex.

[10] Index of Surface Variance: This index shows the rate of deviation of the corneal curvature radius from the mean values of the normal population.

[11] Index of Vertical Asymmetry: Vertical asymmetric index, asymmetry of astigmatism axes.

[12] Keratoconus Index: This index increases in cases of keratoconus. Values greater than 1.07 are considered abnormal.

[13] Central Keratoconus Index: This index increases in cases of central keratoconus. Values greater than 1.03 are considered abnormal.

[14] Index of Height Decentration: This index measures the amount of displacement of points raised versus vertical meridians.

Table 1 Mean values and standard deviations of variables in the dataset

Group	Eyes	Mean ± Standard Deviation (range)			
		Age(years)	Rh (mm)	Rv (mm)	Astig (D)
Normal	130	38.18 ± 12.9 (18–60)	7.77 ± 0.32 (6.36–8.57)	7.58 ± 0.29 (7.01–8.42)	1.41 ± 0.92(0–5)
KC1	120	38.75 ± 11.83 (18–60)	7.71 ± 0.30(6.93–8.44)	7.49 ± 0.31(6.8–8.37)	2.05 ± 0.88 (0.5–4.8)
KC2	105	39.07 ± 11.67 (18–60)	7.38 ± 0.50 (5.38–8.92)	7.08 ± 0.58 (4.87–8.6)	3.301 ± 1.726 (0 to 9.5)
KC3	95	38.87 ± 12.84 (18–60)	6.90 ± 0.54 (5.68–8.5)	6.57 ± 0.60 (5.1–8.34)	3.46 ± 2.12 (0–9.4)

Group	Mean ± Standard Deviation (range)				
	Dp	Dt	Da	ISV	KI
Normal	0.93 ± 2.09(−1.34 to 19.35)	0.26 ± 1.23 (−2.31 to 5.6)	0.63 ± 0.98(−2.94 to - 2.94)	19.37 ± 7.77 (9–44)	1.02 ± 0.02 (0.98–1.12)
KC1	3.66 ± 2.38(−2.23 to 13.58)	1.52 ± 1.29 (−4.38 to 4.51)	2.12 ± 0.97 (−1.61 to 3.49)	36.93 ± 4.50 (27–45)	1.08 ± 0.02 (1.01–1.14)
KC2	8.24 ± 6.54(- 1.15to59.84)	3.27 ± 2.19(-0.5 to 17.77)	2.99 ± 0.50 (0.16–4.35)	70.6 ± 7.61 (58–84)	1.18 ± 0.039 (1–1.26)
KC3	11.90 ± 5.39(4.03–46.92)	4.40 ± 1.79 (0.57–11.06)	3.41 ± 0.31 (2.18–4.16)	114.31 ± 11.46 (100–145)	1.31 ± 0.06 (0.99–1.48)

corneal curvature, eccentricity, anterior chamber, corneal volume, and pachymetry. More details on the KC dataset are presented in Table 1.

Data pre-processing

The data were analyzed using Microsoft Excel and MATLAB software programs to perform learning techniques. For ensuring that no out-of-range values were used for testing and training, using histograms and outlier detection, the dataset was reviewed for unrealistic and improper values. Finally, a normalization method was performed to fit all variables into a determined range and prepare them to be applied to the ensemble system.

The K-fold cross-validation method (CV method) [37] was used to randomly partition the original dataset into K equal-sized subsamples. The dataset was randomly divided into two sections. The first section was used for training and validating the ensemble system. This section constituted 80% of the original data. The remaining data of 20% were used for testing the system. This 20% data were never used for training or validating the system. Thus, the samples were randomly partitioned into five equal-sized subsets; each time one subset was considered as testing data and the other four subsets were considered as training and validation data. Then, the training and validation data (80% of the whole dataset) were divided into 10 equal sections. Across 10 iterations, among 10 subsamples, 9 subsamples were used to train the system, and the remaining single subsample was retained as the validation data. Thus, in each of iterations, 90% of the data were used for training and 10% of them were applied for validating. Measurement accuracy of this system was average of accuracy values of test data measured over these five performances. In addition, 200 iterations were used to train the system.

Evaluation criteria of classification

Efficiency of the ensemble system was determined based on the number of true-negative (tn), true-positive (tp), false-negative (fn), and false-positive (fp) cases. These parameters can range between 0 and 100% (or between 0 and 1) so that, 100% represents the best values and 0% shows the worst values. Since this was a multi-class prediction, the following descriptions were used for accuracy, sensitivity, specificity, precision, and F1_score [38].

Accuracy = It is referred to the ratio of number of correct predictions to the total number of input samples.

$$\textbf{Accuracy} = \frac{(tp + tn)}{\textbf{total number of inputs}} \tag{7}$$

Sensitivity: It is used to determine proportion of actual positive cases, predicted correctly.

$$\textbf{Sensitivity} = \frac{tp}{(tp + fn)} \tag{8}$$

Specificity: It is used to determine proportion of actual negative cases, predicted correctly.

$$\textbf{Specificity} = \frac{tn}{(tn + fp)} \tag{9}$$

Precision: It is mean fraction of actual positive cases among the retrieved instances.

$$\textbf{Precision} = \frac{tp}{(tp + fp)} \tag{10}$$

F1_score: It considers both precision and sensitivity of the system to compute the score.

$$F1\_\textbf{score} = \frac{tp}{\left(tp + \frac{1}{2}(fp + fn)\right)} \tag{11}$$

Results and discussion

For evaluating the ability of the system to precisely classify corneas in the four clinical groups, two classification tasks were performed, namely KC versus *N* and a classification including the four groups. Sensitivity, specificity, precision, F1_score, and accuracy were computed using the CV method.

KC and N eyes

Table 2 presents results of classification between KC and normal groups using two ensemble systems with different basic classifiers and weighted voting combination method (MLP_WV and ANFIS_WV). Results for the same ensemble systems and Naïve Bayes combination method (MLP_NB and ANFIS_NB) are presented in Table 3. The accuracy achieved by the MLP_WV ensemble system was equal to 95.8%, and sensitivity and specificity were equal to 97.8 and 90.1% for KC diagnosis, respectively. Only, 7 KC eyes were misclassified (out of 320). The accuracy achieved by the ANFIS_WV ensemble system was equal to 96.2%, with sensitivity and specificity of 99.1 and 89.2% for KC diagnosis, respectively. There were only 3 misclassified KC eyes (out of 320).

Likewise, the accuracy achieved by the MLP_NB ensemble system was equal to 95.8%, and sensitivity and specificity were equal to 98.1 and 90.0% for KC diagnosis, respectively. There were only 6 misclassified KC eyes (out of 320). The accuracy achieved by the ANFIS_NB ensemble system was equal to 96.7%, and sensitivity and specificity were equal to 96.3 and 97.7% for KC diagnosis, respectively. There were 12 misclassified KC eyes (out of 320).

Four-group classification: KC1, KC2, KC3, and N

The global accuracy of the ensemble systems with MLP and ANFIS initial classifiers and weighted voting combination method that was related to the four groups (*MLP_WV_G* and *ANFIS_WV_G*) was equal to 94.5 and 94.5%, respectively. The *MLP_WV_G* ensemble systems had an average sensitivity of 94.7% and specificity of 98.1% and the *ANFIS_WV_G* ensemble systems had an average sensitivity of 94.8% and specificity of 98.2% (Table 4).

Similarly, the global accuracy for the same ensemble systems and Naïve Bayes combination method (*MLP_NB_G* and *ANFIS_NB_G*) was equal to 91.1 and 88.4%, respectively, with an average sensitivity of 90.9% and specificity of 97.0% for the *MLP_NB_G* ensemble systems, and an average sensitivity of 87.5% and specificity of 96.1% for the *ANFIS_NB_G* ensemble systems (Table 5).

Combination of both MLP and ANFIS

After evaluating the system with MLP and ANFIS classifiers separately, a combination of both classifiers was used for testing the system. In this regard, different combinations of these two initial classifiers were considered for each data category, and the best combination of them was selected based on the system results.

In such a system, neural network was used for groups of Front and D data; ANFIS classifier was used for groups of Back, ISV, and Hybrid data.

Table 2 Confusion Matrix (Actual vs. Predicted Classes), Sensitivity, Specificity, Precision, F1_Score, and CV Accuracy of the ensemble system with MLP and ANFIS initial classifiers alongside *weighted voting* combination method for KC versus N distinction

Ensemble system	Basic classifiers	Actual class		Predicted class		Sensitivity (%)	Specificity (%)	Precision (%)	F1_score (%)	Accuracy (%)
				KC	N					
MLP_WV	MLP	KC	320	313	7	97.8	90.1	96.3	97.1	95.8
		N	130	12	118	90.1	97.8	94.4	92.5	
ANFIS_WV	ANFIS	KC	320	317	3	99.1	89.2	95.8	97.4	96.2
		N	130	14	116	89.2	99.1	97.4	93.2	

Table 3 Confusion Matrix (Actual vs. Predicted Classes), Sensitivity, Specificity, Precision, F1_Score, and CV Accuracy of the ensemble system with MLP and ANFIS initial classifiers alongside Naïve Bayes combination method for KC versus N distinction

Ensemble system	Basic classifiers	Actual class		Predicted class		Sensitivity (%)	Specificity (%)	Precision (%)	F1_score (%)	Accuracy (%)
				KC	N					
MLP_NB	MLP	KC	320	314	6	98.1	90.0	96.0	97.1	95.8
		N	130	13	117	90.0	98.1	95.1	92.5	
ANFIS_NB	ANFIS	KC	320	308	12	96.3	97.7	99.0	97.6	96.7
		N	130	3	127	97.7	96.3	91.4	94.4	

Results of the classification task using such a system with weighted voting and Naïve Bayes combination methods are presented in Tables 6 and 7. The accuracy achieved by the *MLP_ANFIS_WV* and *MLP_ANFIS_NB* ensemble systems was equal to 96.7 and 98.2%, respectively. Sensitivity and specificity were equal to 99.1 and 90.1% for KC diagnosis using the *MLP_ANFIS_WV* system, respectively. Similarly, sensitivity and specificity were equal to 99.1 and 96.2% for KC diagnosis using *MLP_ANFIS_NB* system, respectively. There were only 3 misclassified KC eyes (out of 320).

The global accuracy of such systems with weighted voting and Naïve Bayes combination method (*MLP_ANFIS_WV_G* and *MLP_ANFIS_NB_G*) that was related to the four groups was equal to 95.6 and 98.2%, respectively. The average sensitivity of 95.9% and specificity of 98.5% were achieved by the *MLP_ANFIS_WV_G* ensemble system, and the average sensitivity of 98.4% and specificity of 99.4% were achieved by the *MLP_ANFIS_NB_G* ensemble system.

Discussion

Clinical diagnosis of KC is quantitatively supported by corneal topography. In automatic diagnosis of KC, corneal topographic patterns are very important, especially in preparative examination for refractive surgery.

The results of this study showed that a combination of several classifiers trained separately can improve recognition rate of the ensemble system. In this system, 19 parameters were selected from four data groups of corneal topography (Table 1). One of the important results of this research was that the use of 19 parameters from both sides of the cornea and the use of different initial classifiers increased power of the ensemble system.

In any case, the best results were obtained using the ensemble system with both MLP and ANFIS classifiers and the Naïve Bayes combination method (*MLP_ANFIS_NB*), with 98.2% of accuracy, 99.1% of sensitivity, and 96.2% of specificity, which were slightly better than the ensemble systems with

Table 4 Confusion Matrix (Actual vs. Predicted Classes), Sensitivity, Specificity, Precision, F1_Score, and CV Accuracy of the ensemble system with MLP and ANFIS initial classifiers alongside weighted voting combination method for 4 Groups classification

Ensemble system	Basic classifiers	Actual class		Predicted class				Sensitivity (%)	Specificity (%)	Precision (%)	F1_Score (%)	Accuracy (%)
				KC1	KC2	KC3	N					
		KC1	120	114	0	0	6	95.00	96.4	90.5	92.7	
MLP_WV_G	MLP	KC2	105	0	102	2	1	97.1	98.8	96.5	96.7	94.5
		KC3	95	0	4	91	0	95.8	99.4	97.8	96.7	
		N	130	12	0	0	118	90.8	97.8	94.5	92.7	
		KC1	120	116	1	0	3	96.7	95.2	88.0	92.1	
ANFIS _WV_G	ANFIS	KC2	105	1	99	5	0	94.3	99.7	98.8	96.4	94.5
		KC3	95	1	0	94	0	98.9	98.6	95.4	97.0	
		N	130	14	0	0	116	89.2	99.1	97.5	93.2	

Table 5 Confusion Matrix (Actual vs. Predicted Classes), Sensitivity, Specificity, Precision, F1_Score, and CV Accuracy of the ensemble system with MLP and ANFIS initial classifiers alongside Naïve Bayes combination method for 4 Groups classification

Ensemble system	Basic classifiers	Actual class		Predicted class				Sensitivity (%)	Specificity (%)	Precision (%)	F1_Score (%)	Accuracy (%)
				KC1	KC2	KC3	N					
		KC1	120	114	0	0	6	95.0	89.7	77.2	85.1	
MLP_NB_G	MLP	KC2	105	7	98	0	0	93.3	100	100	96.5	91.1
		KC3	95	14	0	81	0	85.3	100	100	92.0	
		N	130	13	0	0	117	90.0	98.1	95.1	92.5	
		KC1	120	108	0	0	12	90.0	87.9	73.2	80.5	
ANFIS _NB_G	ANFIS	KC2	105	11	94	0	0	89.5	100	100	94.4	88.4
		KC3	95	26	0	69	0	72.6	100	100	84.1	
		N	130	3	0	0	127	97.7	96.3	91.8	94.5	

weighted voting combination method (*MLP_ANFIS_WV*), with 96.7% of accuracy, 99.1% of sensitivity, and 90.1% of specificity (see Table 6).

In fact, a large number of parameters can be obtained from topographic maps and can be used in various automatic methods to identify patterns of corneal pathologies. Several studies in the literature have successfully distinguished between keratoconic and normal corneas using different quantitative parameters and machine learning algorithms. For instance, Hidalgo et al. [39] in a study, assessed performance of a support vector machine (SVM) for KC diagnosis and achieved the accuracy of 98.9%, with 99.1% of sensitivity and 98.5% of specificity for subclinical KC diagnosis. These achievements were almost similar to the values achieved in our study (98.2% of accuracy, 99.1% of sensitivity, and 96.2% of specificity).

However, although each stage of KC requires its different specific treatment steps and must be diagnosed separately, they considered all the stages of KC belonging to one category (KC) and did not distinguish between them. For example, a sample with KC stage 3 may be misclassified as stage one (see Table 1), but since it is still part of the KC category, it is considered a correct detection. In our study, each stage of KC was regarded as a distinct group. In this study, for 4-group classification, the best result was achieved by the *MLP_ANFIS_NB* ensemble system (98.2% of accuracy, with an average sensitivity of 98.4% and specificity of 98.8%). Furthermore, they used a combination of 22 parameters obtained from Pentacam measurements, while herein, 19 parameters were applied.

Arbelaez et al. in another study used SVM for distinguishing between KC and N corneas [40]. Efficiency of a SVM for KC diagnosis was evaluated both with and without the data generated from corneal thickness and posterior corneal surface. They classified eyes into four groups: abnormal, KC, subclinical KC, and normal. Applying the data from both posterior and anterior corneal surfaces and pachymetry increased sensitivity of the SVM from 93.1 to 97.2% in normal eyes, 75.2 to 92.0% in eyes with subclinical KC, 92.8 to 95.0% in eyes with KC, and 89.3 to 96.0% in abnormal eyes. Accordingly, in this study, the data from anterior and posterior corneal surfaces were used and the accuracy of 98.2% was achieved with sensitivity of 99.1% for KC diagnosis.

Accardo et al. [41] described a neural network-based system. Using a videokeratoscope, they selected 396 corneal maps from the obtained corneal maps. At first, the data taken from each eye were considered separately. Then, they used the parameters evaluated from both eyes. They divided the maps into KC, N, and other non-KC conditions (O). However, they considered only the early KC eyes (KC1). The best results were achieved using the parameters of both eyes of the same subject as input. In this condition, they obtained a global sensitivity and specificity of 94.1and 97.6%, respectively.

Toutounchian et al. [42] focused on using Decision Tree, SVM, radial basis function (RBF) neural network, and MLP neural network to distinguish suspected cases of KC from definite KC cases. They classified eyes into three categories: normal, suspected KC cases, and definite KC cases. Like the previous studies, they considered all KC eyes in one category. Results of their study showed that Decision Tree gave the best result (84% of accuracy) and RBFNN as the worst tool yielded 71.20% of accuracy. Generally, the accuracy of about 91% was achieved for detection of normal eye, suspected KC cases, and definite KC cases by the proposed algorithm.

Table 6 Confusion Matrix (Actual vs. Predicted Classes), Sensitivity, Specificity, Precision, F1_Score, and CV Accuracy of the ensemble system with the combination of both MLP and ANFIS classifiers alongside weighted voting and naïve Bayes combination methods for KC versus N distinction

Ensemble system	Basic classifiers	Combination method	Actual class		Predicted class		Sensitivity (%)	Specificity (%)	Precision (%)	F1_Score (%)	Accuracy (%)
					KC	N					
MLP_ANFIS_WV	MLP & ANFIS	Weighted voting	KC	320	317	3	99.1	90.1	96.3	97.7	96.7
			N	130	12	118	90.1	99.1	97.5	94.0	
MLP_ANFIS_NB	MLP & ANFIS	Naïve Bayes	KC	320	317	3	99.1	96.2	98.4	98.8	98.2
			N	130	5	125	96.2	99.1	97.7	96.9	

Table 7 Confusion Matrix (Actual vs. Predicted Classes), Sensitivity, Specificity, Precision, F1_Score, and CV Accuracy of the ensemble system with the combination of both MLP and ANFIS classifiers alongside weighted voting and naïve Bayes combination methods for 4 Groups classification

Ensemble system	Basic classifiers	Combination method	Actual class		Predicted class				Sensitivity (%)	Specificity (%)	Precision (%)	F1_Score (%)	Accuracy (%)
					KC1	KC2	KC3	N					
			KC1	120	116	1	0	3	96.67	96.1	90.0	93.2	
MLP_ANFIS_WV_G	MLP and ANFIS	Weighted voting	KC2	105	1	101	3	0	96.19	99.7	99.0	97.6	95.6
			KC3	95	0	0	95	0	100	99.2	97.0	98.5	
			N	130	12	0	0	118	90.77	99.1	97.6	94.0	
			KC1	120	117	0	0	3	97.5	98.5	95.9	96.7	
MLP_ANFIS_NB_G	MLP & ANFIS	Naïve Bayes	KC2	105	0	105	0	0	100	100	1	1	98.2
			KC3	95	0	0	95	0	100	100	1	1	
			N	130	5	0	0	125	96.2	99.1	97.8	96.9	

Smadja et al. [43] presented a method for diagnosis of subclinical KC based on a Decision Tree. They used a dataset from dual–Scheimpflug tomography. They analyzed 55 parameters obtained from posterior and anterior corneal measurements for each eye and classified the eyes into N, forme fruste KC (FFKC), and KC. Discriminatory rules generated by the Decision Tree classifier allowed detection of normal and KC eyes with 100% of sensitivity, and 99.5% of specificity. The sensitivity of 93.6% and specificity of 97.2% were achieved for detection of normal and FFKC eyes. In our study, only 19 parameters(less than half of the number of parameters used in the mentioned study) were used and 99.1% of sensitivity and 96.2% of specificity were obtained for KC diagnosis.

Conclusion

In conclusion, in the current study, a new ensemble system was introduced for solving classification problems related to uncertainty in diagnosis of KC. Despite using fewer parameters, herein, comparable or, in some cases, better results were obtained than methods reported in the literature. In some cases, it was not possible to directly compare our results with the literature, due to differences in definitions of KC group as well as differences in selection of items and parameters. The proposed system can be used in the Pentacam as a utility program to aid the ophthalmologist through quick screening of patients. However, it should not be considered as a stand-alone diagnostic tool but it can be applied as an additional tool alongside other clinical data. Although this method demonstrated a very good accuracy in discriminating between normal eyes and the eyes with different stages of KC, it is hoped to be able to focus on subclinical and forme fruste keratoconus in future works and detect asymptomatic subclinical patients as well as corneas suspicious to KC. Moreover, it is possible and valuable for future works to develop the proposed model and adapt it to more recent AS-OCT devices and other Scheimpflug systems.

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Author contributions MG and AS designed the model and computational framework. They also analyzed the data, developed the theoretical formulation, and performed the analytic calculations and the numerical simulations. They designed the figures for the suggested experiment as well. AS and EJP supervised this study in the concepts of computer techniques and ophthalmic concepts, respectively. All authors aided in interpreting the results, provided critical feedback, discussed the results, and contributed to the final version of manuscript. All authors approved the final version of the submitted manuscript.

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Availability of data and material Clinical data and corneal examinations were retrieved from clinical records at the Noor Medical Eye Clinic. The dataset consists of information from both the anterior and posterior corneal surfaces of 450 eyes. The eyes, which were studied in this research, were divided into four classes (Normal, KC1, KC2, KC3) based on the cornea center topography results.

Code availability Due to the fact that this research and collaboration with Noor Center is ongoing, the Code is currently not publicly available.

Declarations

Conflict of interest The authors certify that they have NO affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in the subject matter or materials discussed in this manuscript. The authors affirmed that this manuscript is original, has not been published before, and is not currently being considered for publication elsewhere.

Ethical approval This research study was conducted retrospectively from data obtained for clinical purposes. We consulted with the ethics committee of the Noor Medical Eye Clinic, and they determined that our study did not need ethical approval.

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